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Editor-in-Chief
Applied Sciences
MDPI

Dear Editor,

I am pleased to submit our manuscript entitled “Temporal Attention-Enhanced Lightweight CNN for Student Engagement Detection on Edge Devices” for consideration for publication in *Applied Sciences*.

This paper proposes a novel lightweight framework for real-time student engagement detection on edge devices, with three key contributions:

1. A multi-scale Channel-Spatial Attention Module (CSAM) with parallel 3×3 , 5×5 , 7×7 convolutions and adaptive fusion, designed for mobile architectures.
2. A BiLSTM-based temporal aggregation module that converts frame-level predictions to video-level engagement classification.
3. Comprehensive evaluation on three datasets with deployment-oriented metrics (parameters, FLOPs, latency, energy).

Key results demonstrate the effectiveness of temporal modeling:

- FER2013: 70.5% accuracy (5-fold CV, 1.26M parameters)
- CK+ cross-dataset: 71.9% accuracy (without fine-tuning)
- DAiSEE: 94.8% video-level accuracy (vs 61.2% frame-level)

The dramatic improvement from frame-level to video-level ($61.2\% \rightarrow 94.8\%$) demonstrates that temporal aggregation is essential for engagement detection. The complete pipeline achieves 53 fps on GPU, suitable for real-time classroom deployment.

Both authors have read and approved the manuscript. The work has not been published elsewhere and is not under consideration by another journal. We declare no conflicts of interest.

I look forward to your decision.

Sincerely,

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